



COLLEGE OF ENGINEERING & TECHNOLOGY

Department: Basic and Applied Sciences

EBA0201 – PRECALCULUS

Lecture 1

LIMITS

$$\lim_{x \rightarrow a} f(x)$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$



Build
Stronger Foundations
for a Brighter Tomorrow

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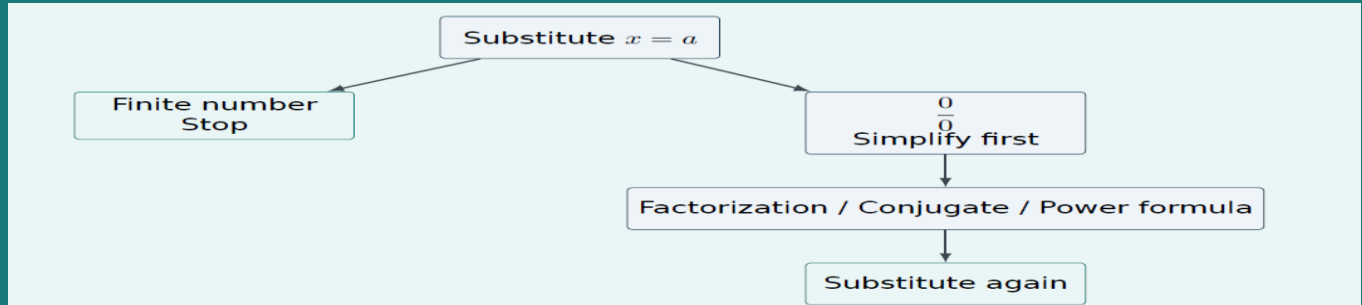


1. The First Rule: Substitute First

Basic rule

For an elementary function $f(x)$, start by direct substitution: $\lim_{x \rightarrow a} f(x) = f(a)$

The decision path



Important

0/0 is an indeterminate form. It is not the answer. It tells us that the expression must be simplified before substituting again.

Example 1

Evaluate

$$\lim_{x \rightarrow -1} \sqrt{x^2 - 3x + 2}$$

Solve:

$$\lim_{x \rightarrow -1} \sqrt{x^2 - 3x + 2} = \sqrt{(-1)^2 - 3(-1) + 2} = \sqrt{1 + 3 + 2} = \boxed{\sqrt{6}}$$

Example 2

Evaluate

$$\lim_{x \rightarrow 2} (x^3 - 5x^2 + 7)$$

solve:

$$\lim_{x \rightarrow 2} (x^3 - 5x^2 + 7) = 2^3 - 5(2)^2 + 7 = 8 - 20 + 7 = \boxed{-5}$$

Example 3

Evaluate

$$\lim_{x \rightarrow 3} (2x^2 - 4x + 1)$$

solve:

$$\lim_{x \rightarrow 3} (2x^2 - 4x + 1) = 2(3)^2 - 4(3) + 1 = \boxed{7}$$

Example 4

Evaluate

$$\lim_{x \rightarrow 4} \sqrt{x + 5}$$

solve:

$$\lim_{x \rightarrow 4} \sqrt{x + 5} = \sqrt{9} = \boxed{3}$$

Example 5 - Medium

Evaluate

$$\lim_{x \rightarrow -2} \frac{x^2 + 3}{x - 1}$$

solve:

$$\lim_{x \rightarrow -2} \frac{x^2 + 3}{x - 1} = \frac{4 + 3}{-3} = \boxed{-\frac{7}{3}}$$

2. Limits by Factorization

When do we factor?

If direct substitution gives $0/0$ and the expression contains polynomials, factor the numerator and/or denominator, cancel the common factor, then substitute again.

Substitute $\rightarrow \frac{0}{0} \rightarrow$ Factor $\rightarrow$ Cancel $\rightarrow$ Substitute again

Example 6

Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{2x^2 - x - 6}$$

solve:

Direct substitution gives $0/0$.

$$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{2x^2 - x - 6} = \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(x-2)(2x+3)} = \lim_{x \rightarrow 2} \frac{x+1}{2x+3} = \boxed{\frac{3}{7}}$$

Example 7

Evaluate

$$\lim_{x \rightarrow 0} \frac{5x}{x^2 + 3x}$$

solve:

Direct substitution gives $0/0$.

$$\lim_{x \rightarrow 0} \frac{5x}{x^2 + 3x} = \lim_{x \rightarrow 0} \frac{5x}{x(x+3)} = \lim_{x \rightarrow 0} \frac{5}{x+3} = \boxed{\frac{5}{3}}$$

Example 8

Evaluate

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x^2 - 7x + 12}$$

solve:

Direct substitution gives $0/0$.

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x^2 - 7x + 12} = \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{(x-4)(x-3)} = \lim_{x \rightarrow 4} \frac{x+4}{x-3} = \boxed{8}$$

Example 9

Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x^3 + x^2 - 2x - 8}$$

solve:

Direct substitution gives $0/0$. Factor the numerator:

$$x^2 + 3x - 10 = (x-2)(x+5)$$

For the denominator, group the terms:

$$\begin{aligned} x^3 + x^2 - 2x - 8 &= (x^3 - 8) + (x^2 - 2x) \\ &= (x-2)(x^2 + 2x + 4) + x(x-2) \\ &= (x-2)(x^2 + 3x + 4) \end{aligned}$$

Thus

$$\lim_{x \rightarrow 2} \frac{x+5}{x^2 + 3x + 4} = \frac{7}{14} = \boxed{\frac{1}{2}}$$

Example 10 - Easy

Evaluate

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

solve:

Direct substitution gives 0/0.

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} = \boxed{6}$$

Example 11 - Medium

Evaluate

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$$

solve:

Direct substitution gives 0/0.

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)} = \frac{12}{4} = \boxed{3}$$

Example 12

Evaluate

$$\lim_{x \rightarrow 1} \frac{x^3 - 3x + 2}{x^2 - 2x + 1}$$

solve:

Direct substitution gives 0/0. Factor both numerator and denominator:

Therefore

$$x^3 - 3x + 2 = (x - 1)^2(x + 2), \quad x^2 - 2x + 1 = (x - 1)^2$$

$$\lim_{x \rightarrow 1} \frac{(x - 1)^2(x + 2)}{(x - 1)^2} = \lim_{x \rightarrow 1} (x + 2) = \boxed{3}$$

3. Limits Using the Conjugate

When do we use the conjugate?

When radicals are present and direct substitution gives 0/0, multiply by the conjugate.

$$(\sqrt{A} - B)(\sqrt{A} + B) = A - B^2$$

Example 13

Evaluate

$$\lim_{x \rightarrow 7} \frac{\sqrt{x - 3} - 2}{x^2 - 49}$$

solve:

Direct substitution gives 0/0. Multiply by the conjugate:

$$\lim_{x \rightarrow 7} \frac{\sqrt{x - 3} - 2}{x^2 - 49} \times \frac{\sqrt{x - 3} + 2}{\sqrt{x - 3} + 2} = \lim_{x \rightarrow 7} \frac{x - 7}{(x^2 - 49)(\sqrt{x - 3} + 2)} = \lim_{x \rightarrow 7} \frac{1}{(x + 7)(\sqrt{x - 3} + 2)} = \boxed{\frac{1}{56}}$$

Example 14

Evaluate

$$\lim_{x \rightarrow 1} \frac{\sqrt{x + 3} - 2}{x - 1}$$

solve:

Direct substitution gives 0/0. Multiply by the conjugate:

$$\lim_{x \rightarrow 1} \frac{\sqrt{x + 3} - 2}{x - 1} \times \frac{\sqrt{x + 3} + 2}{\sqrt{x + 3} + 2} = \lim_{x \rightarrow 1} \frac{x - 1}{(x - 1)(\sqrt{x + 3} + 2)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x + 3} + 2} = \boxed{\frac{1}{4}}$$

Example 15

Evaluate

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$$

solve:

Direct substitution gives 0/0. Multiply by the conjugate:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} \times \frac{\sqrt{x} + 3}{\sqrt{x} + 3} = \lim_{x \rightarrow 9} \frac{1}{\sqrt{x} + 3} = \boxed{\frac{1}{6}}$$

Example 16

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

solve:

Direct substitution gives 0/0. Multiply by the conjugate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} \times \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} = \boxed{\frac{1}{4}}$$

Example 17

Evaluate

$$\lim_{x \rightarrow 5} \frac{x - 5}{\sqrt{x+4} - 3}$$

solve:

Direct substitution gives 0/0. Multiply by the conjugate

$$\lim_{x \rightarrow 5} \frac{x - 5}{\sqrt{x+4} - 3} = \lim_{x \rightarrow 5} \frac{(x - 5)(\sqrt{x+4} + 3)}{x + 4 - 9} = \lim_{x \rightarrow 5} (\sqrt{x+4} + 3) = \boxed{6}$$

4. A Useful Power Formula

Special form

For limits of the form

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{0}{0}$$

$$\boxed{\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}}$$

Example 18

Evaluate

$$\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^4 - 16}$$

solve:

Direct substitution gives 0/0.

$$\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^4 - 16} = \frac{5}{4} (2)^{5-4} = \boxed{\frac{5}{2}}$$

Example 19

Evaluate

$$\lim_{x \rightarrow 2} \frac{5x^7 - 640}{x^4 - 16}$$

solve:

Direct substitution gives 0/0.

$$\lim_{x \rightarrow 2} \frac{5x^7 - 640}{x^4 - 16} = 5 \lim_{x \rightarrow 2} \frac{x^7 - 2^7}{x^4 - 2^4} = 5 \left(\frac{7}{4}\right) 2^3 = \boxed{70}$$

Example 20

Evaluate

$$\lim_{x \rightarrow \frac{1}{2}} \frac{x^2 - \frac{1}{4}}{4x^5 - \frac{1}{8}}$$

solve:Direct substitution gives 0/0. Since $4x^5 - \frac{1}{8} = 4\left(x^5 - \frac{1}{32}\right)$,

$$\lim_{x \rightarrow \frac{1}{2}} \frac{x^2 - \frac{1}{4}}{4x^5 - \frac{1}{8}} = \lim_{x \rightarrow \frac{1}{2}} \frac{x^2 - \left(\frac{1}{2}\right)^2}{4\left[x^5 - \left(\frac{1}{2}\right)^5\right]} = \frac{1}{4} \left(\frac{2}{5}\right) \left(\frac{1}{2}\right)^{-3} = \boxed{\frac{4}{5}}$$

Example 21

Evaluate

$$\lim_{x \rightarrow 64} \frac{\sqrt{x} - 8}{\sqrt[3]{x} - 4}$$

solve:Direct substitution gives 0/0. Write the powers as $x^{\frac{1}{2}}$ and $x^{\frac{1}{3}}$

$$\lim_{x \rightarrow 64} \frac{\sqrt{x} - 8}{\sqrt[3]{x} - 4} = \frac{1}{2} (64)^{\frac{1}{2} - \frac{1}{3}} = \frac{3}{2} (64)^{1/6} = \frac{3}{2} (2) = \boxed{3}$$

Example 22

Evaluate

$$\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^3 - 8}$$

solve:

Direct substitution gives 0/0.

$$\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^3 - 8} = \frac{6}{3} (2)^3 = \boxed{16}$$

Example 23

Evaluate

$$\lim_{x \rightarrow -3} \frac{x^5 + 243}{x^4 - 81}$$

solve:

Direct substitution gives 0/0.

$$\lim_{x \rightarrow -3} \frac{x^5 + 243}{x^4 - 81} = \frac{5}{4} (-3)^1 = \boxed{-\frac{15}{4}}$$

5. Quick Decision Guide

Choose the method from the form

What you see	First action	Typical method
A polynomial, radical, or rational function with a defined value	Substitute directly	Direct substitution
0/0 with polynomial factors	Simplify	Factorization and cancellation
0/0 with square roots	Remove the radical difference	Multiply by the conjugate
$\frac{x^n - a^n}{x^m - a^m}$	Recognize the power pattern	Power formula or factorization

6. Class Practice

Level 1 - Direct substitution

1. $\lim_{x \rightarrow 2} (x^2 + 3x - 1)$
2. $\lim_{x \rightarrow 1} \frac{3x + 4}{x + 5}$
3. $\lim_{x \rightarrow 4} \sqrt{x + 12}$

Level 2 - Factorization

4. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$
5. $\lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x - 2}$
6. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$

Level 3 - Conjugate and powers

7. $\lim_{x \rightarrow 0} \frac{\sqrt{x + 9} - 3}{x}$
8. $\lim_{x \rightarrow 3} \frac{x - 3}{\sqrt{x + 1} - 2}$
9. $\lim_{x \rightarrow 2} \frac{x^7 - 128}{x - 2}$
10. $\lim_{x \rightarrow -4} \frac{x^3 + 64}{x + 4}$